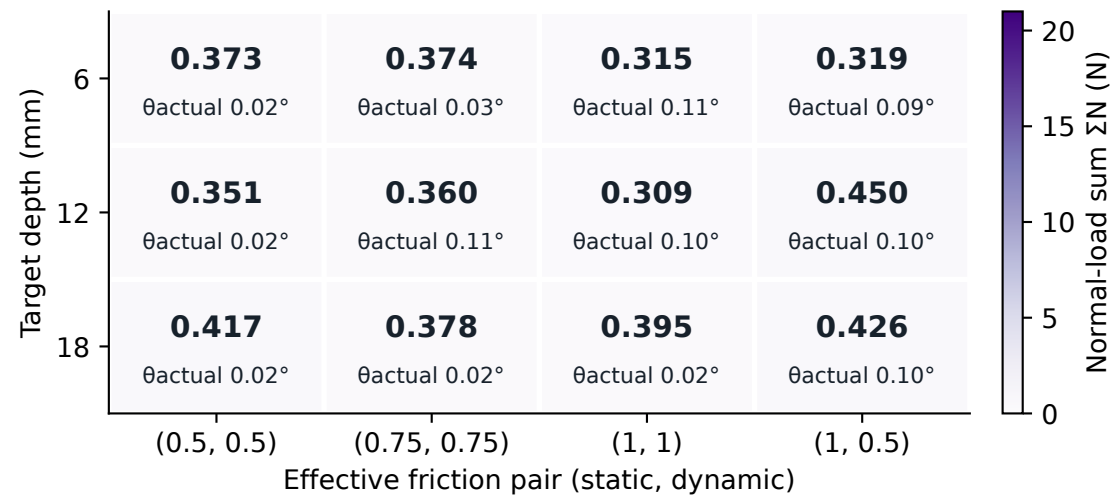
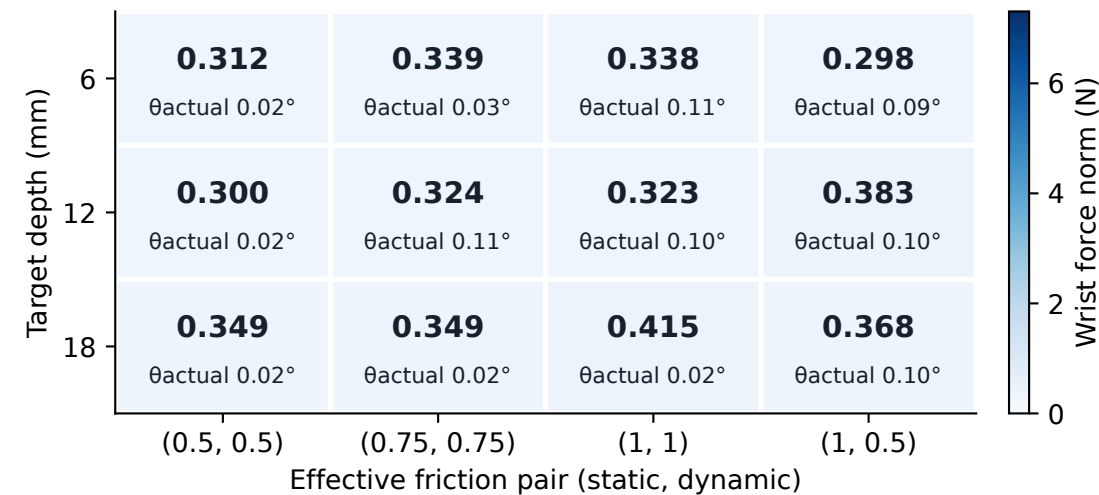


Reference contact loads and wrist reaction
24 closed attempts / 24 planned conditions; hatched cells are truncated references

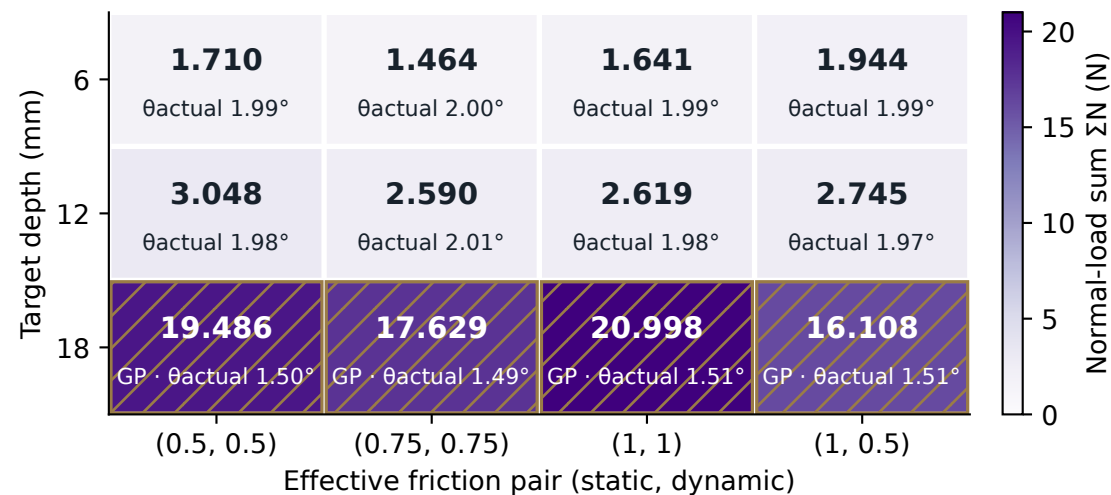
Target tilt 0° | Peak Σ contact-normal magnitudes



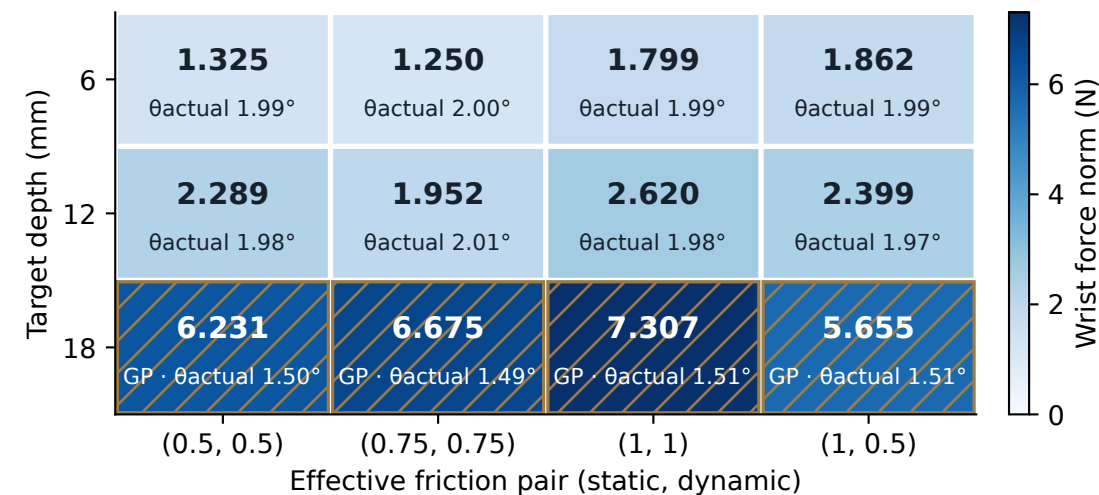
Target tilt 0° | Peak wrist force norm



Target tilt 2° | Peak Σ contact-normal magnitudes



Target tilt 2° | Peak wrist force norm



Independent color scales: ΣN sums contact-normal magnitudes, including opposing sides. ΣN is neither axial pull force nor wrist-force norm.

Cells show the recorded reference peak and actual terminal angle. Hatching + GP/BD/DG/TI retains observed data from a reference stopped before full completion.

GP = grasp-slip threshold reached (not demonstrated dropping); BD = operating budget; DG = depth gate not reached; TI = tilt sequence incomplete.

Grey masks: — pending; RUN active/incomplete file; EX excluded; NV numerical reject; NA missing; N/A unplanned. No missing or rejected peak is replaced by zero.

These are reference-phase loads, not withdrawal costs. One run per condition; descriptive evidence only.